

AE-402: Radiation of [Ru(CO)(OH)₂(PNP)] in water with liquid phase O₂ measurment

Date: 2024-12-09

Tags: Radiation O₂ PNP

[Ru(CO)(OH)₂(PNP)] AE liquidphase

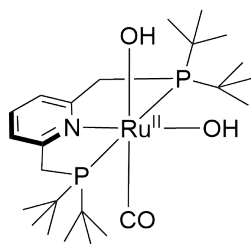
Category: Two-photon

Status: Done

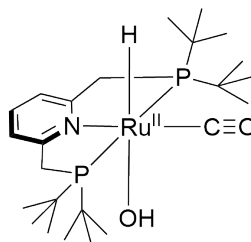
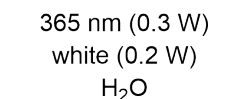
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP^tBu)]



Chemical Formula: C₂₄H₄₅NO₂P₂Ru
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^tBu)]

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-358: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH Two-photon - AE-386: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D ₂ O I

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	Two-photon - AE-382: Synthesis of [Ru(CO)(OH) ₂ (PNP ^t Bu)] using Ag ₂ O, 5 h (AE-382-3)	4.90	/	/	2.74	558.65	/
Milli-Q water	GD-018: Redegassing of H ₂ O	/	/	/	/	/	1

Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9
Thermostat temperature [°C]	25
Stirring speed [rpm]	1000
Irradiation start [s]	2130
Irradiation stop [s]	10320

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
09.12		A 10 mL screw cap schlenk flask was prepared	
	10:40	To the flask [Ru] was added under air	
		The flas was set under argon and stored in Equipment - Freezer Lab E004 CEEC II under Ar in the dark	
11.12		A screw cap 5 mL irradiation flask was prepared	
	12:50	water was added to the flask	
	12:55 - 13:10	The suspension was stirred	

13:10	The suspension was filtered into the 5 mL flask using a PA syringe filter (pore diameter = 0.2 µm	in flask, after filt.jpg yellowish solution
12:55 - 13:00	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	
13:12	The Robust oxygen probe was attached to the flask using a bola fitting (diameter = 3 mm)	
13:13	The flask was moved to the setup	
13:22	The logging was started (AE-402-1-Ch2)	
13:24 - 14:08	The Schlenk was degassed according to Protocol - Degassing and O2 Measurement with a sensor cap with air-stable substances in the Photochem. setup	
14:08	The degassing stopped by removing the PTFE cannula according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	during degassing.jpg no change c(O2) = approx. -0.1 µmol/L
14:10	The Log (-1-Ch2) was stopped	2024-12-11_132235_AE-402-1-Ch2.txt AE-402-1.png
14:10	The logging was started (AE-402-2-Ch2)	
14:22	The Log (-2-Ch2) was stopped	c(O2) = approx. 0.44 µmol/L, increased after stopping of degassing 2024-12-11_141058_AE-402-2-Ch2.txt AE-402-2.png
14:22	The logging was started (AE-402-3-Ch2)	
14:22 - 14:57	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 2130 s
14:57	The light sources were turned on	
17:13	The light source was turned off	Irradiation duration: 8190 s
17:13 - 17:23	The sample was allowed to equilibrate according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	Equilibration duration: 555 s
16:10	The logging was stopped (-3-Ch2)	after irrad.jpg intenser orange solution 2024-12-11_142207_AE-402-3-Ch2.txt

Analysis

Date	Time	Sample name	Analysis method	Analytic device	Solvent	Raw Data	Processed Data	Interpretation
11.12	13:22	AE-402-1-Ch1	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	2024-12-11_132235_AE-402-1-Ch2.txt	AE-402-1.png	Degassing worked.
11.12	14:10	AE-402-2-Ch1	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	2024-12-11_141058_AE-402-2-Ch2.txt	AE-402-2.png	After stopping of degassing O2 content increased to approx. 0.4 μ M
11.12	14:22	AE-402-3-Ch1	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	2024-12-11_142207_AE-402-3-Ch2.txt	AE-402-3.png	Stable baseline before and after irradiation. At start of irradiation fast drop from 0.43 μ M to 0.05 μ M. Afterwards increase over approx. 5 to 10 min to 0.07 μ M. Than slow and steady decline to -0.03 μ M. No significant change when light was turned off. Temperature mostly as expected. Only slight increase of 0.5 $^{\circ}$ C. After approx. 30 min of irrad drop of T by approx. 0.08 $^{\circ}$ C, no idea way.

Result

No clear result.

Drop at start of irrad was seen in experiments of MLB before for other systems, when residual amounts of O2 remained.

Afterwards looks like O2 formation, but not sure

Linked experiments

- AE-236: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA

- AE-237: Irradiation of PhPDA (AE-234), 1.5 mg/mg SDS, 2 mg/mL PhPDA I

- AE-245: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-249: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-254: Irradiation of PhPDA (AE-245), 1.5 mg/mg SDS, 2 mg/mL PhPDA reproduction
- AE-257: Synthesis of poly(1,4-diphenylbutadiyne)
- AE-258: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-259: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 1 d
- AE-260: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 2 d
- AE-262: Irradiation of PhPDA (AE-257), 1.5 mg/mg SDS, 2 mg/mL PhPDA after 3 d
- AE-276: Synthesis of poly(1,4-diphenylbutadiyne), 1.2 g scale
- AE-279: Irradiation of PhPDA (AE-276-2), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-282: Irradiation of PhPDA (AE-276-1), 1.5 mg/mg SDS, 2 mg/mL PhPDA
- GD-018: Redegassing of H₂O
- AE-313: Synthesis of PPy via Fenton reaction with FeSO₄ and 10 eq. H₂O₂ I
- AE-332: Irradiation of PPy (AE-312), 2 mg/mL
- AE-333: Irradiation of PPy (AE-313), 2 mg/mL
- C3N₄ - AE-398: Irradiation of C3N₄-Co-0.2 from Lingli Ni
- C3N₄ - AE-399: Irradiation of C3N₄-Co-0.4 from Lingli Ni I
- Two-photon - AE-358: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH
- Two-photon - AE-382: Synthesis of [Ru(CO)(OH)₂(PNPtBu)] using Ag₂O, 5 h
- Two-photon - AE-386: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-382) 365 nm and white, 50 mM KOH in D₂O I

Linked resources

Archived - [Degassing procedure for photocatalytic water splitting by polymeric photocatalysts](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Beam Combiner - [Beam Combiner LCS-BC25-0409 -2](#)

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 1\)](#)

Equipment - [Freezer Lab E004 CEEC II](#)

Light Source - [UHP LED 405 nm](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Light Source - [HP LED 405 -2](#)

Light Source - [HP LED 6500 K \(white\) -2](#)

Protocol - [Irradiation setup](#)

Protocol - [UV vis measurements \(BioTek PowerWave HT\)](#)

Protocol - [Determination of H2O2 by Spectroquant](#)

Protocol - [Degassing and O2 Measurement with sensor cap with air stable substances](#)

Protocol - [Determination of H2O2 in the suspensions of photocatalysts using H2O2 test strips](#)

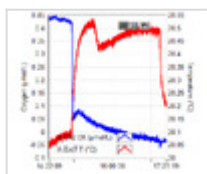
Protocol - [Ultrasonication with the UP 200st from hilscher in ZAF 208](#)

Protocol - [H2O2 photometric detection with DPD-POD method for irradiated suspensions of photocatalysts](#)

Attached files

AE-402-3.png

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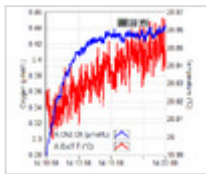


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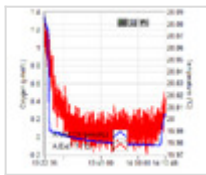


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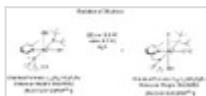


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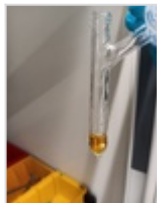


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after-irrad.jpg

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during-degassing.jpg

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in-flask-after-filt.jpg

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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1576>